

● HARD

● CRAFT AND STRUCTURE

Craft and Structure

10 questions · ~15 min

01

Q1

CRAFT AND STRUCTURE

Dubautia carrii is a species in a family of plants known collectively as the silversword alliance, all of which grow only on the Hawaiian Islands. Members of this alliance exhibit an extraordinary range of phenotypes, with some species maturing into vines and others into shrubs and trees. All species in the alliance descended from a single ancestral tarweed plant that arrived on the islands around 5 million years ago. The tarweed's descendants diversified into distinct species as they adapted to live in the wide variety of habitats found on the Hawaiian Islands.

Which choice best states the main purpose of the text?

- A. To indicate the specific tarweed ancestor of all plants that grow on the Hawaiian Islands and explain why the plants have such varied physical characteristics
- B. To describe the specific habitat where *Dubautia carrii* are found and identify other plants that share a common ancestor with them
- C. To describe the silversword alliance and explain how the plant family became so varied
- D. To advance the claim that all plants on the Hawaiian Islands are part of the silversword alliance and list possible ancestors of the plants that make up the alliance

As an undergraduate researcher in anthropology, Jennifer C. Chen contributed to a groundbreaking study challenging the accepted view that among prehistoric peoples, female participation in hunting was _____ blank. The research team's review of data from late Pleistocene and early Holocene burials in the Americas revealed that, in fact, as many as half of the hunters in those populations were female.

Which choice completes the text with the most logical and precise word or phrase?

- A. inevitable
- B. satisfactory
- C. negligible
- D. commonplace

Text 1

Ecologists have long wondered how thousands of microscopic phytoplankton species can live together near ocean surfaces competing for the same resources. According to conventional wisdom, one species should emerge after outcompeting the rest. So why do so many species remain? Ecologists' many efforts to explain this phenomenon still haven't uncovered a satisfactory explanation.

Text 2

Ecologist Michael Behrenfeld and colleagues have connected phytoplankton's diversity to their microscopic size. Because these organisms are so tiny, they are spaced relatively far apart from each other in ocean water and, moreover, experience that water as a relatively dense substance. This in turn makes it hard for them to move around and interact with one another. Therefore, says Behrenfeld's team, direct competition among phytoplankton probably happens much less than previously thought.

Based on the texts, how would Behrenfeld and colleagues (Text 2) most likely respond to the "conventional wisdom" discussed in Text 1?

- A. By arguing that it is based on a misconception about phytoplankton species competing with one another
- B. By asserting that it fails to recognize that routine replenishment of ocean nutrients prevents competition between phytoplankton species
- C. By suggesting that their own findings help clarify how phytoplankton species are able to compete with larger organisms

- D. By recommending that more ecologists focus their research on how competition among phytoplankton species is increased with water density

In Jane Austen’s novel *Mansfield Park*, an almost imperceptible smile from potential suitor Henry Crawford causes the protagonist Fanny Price to blush; her embarrassment grows when she suspects that he is aware of it. This moment—in which Fanny not only infers Henry’s mental state through his gestures, but also infers that *he* is drawing inferences about *her* mental state—illustrates what literary scholar George Butte calls “deep intersubjectivity,” a technique for representing interactions between consciousnesses through which Austen’s novels derive much of their social and psychological drama.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It states a claim about Austen’s skill at representing psychological complexity that is reinforced by an example presented in the following sentence.
- B. It advances an interpretation of an Austen protagonist who is contrasted with protagonists from other Austen novels cited in the following sentence.
- C. It describes a recurring theme in Austen’s novels that is the focus of a literary scholar’s analysis summarized in the following sentence.
- D. It provides a synopsis of an interaction in an Austen novel that illustrates a literary concept discussed in the following sentence.

Political blogs with conspicuous ideological alignments became an integral component of US media in the early 2000s. While some commentators lauded this development, asserting that such blogs had a welcome transparency missing from traditional news, less _____ blank observers countered that such blogs tended to ideological extremes that exacerbated political polarization to problematic levels.

Which choice completes the text with the most logical and precise word or phrase?

- A. sanguine
- B. recalcitrant
- C. misanthropic
- D. earnest

Text 1

Astronomer Mark Holland and colleagues examined four white dwarfs—small, dense remnants of past stars—in order to determine the composition of exoplanets that used to orbit those stars. Studying wavelengths of light in the white dwarf atmospheres, the team reported that traces of elements such as lithium and sodium support the presence of exoplanets with continental crusts similar to Earth’s.

Text 2

Past studies of white dwarf atmospheres have concluded that certain exoplanets had continental crusts. Geologist Keith Putirka and astronomer Siyi Xu argue that those studies unduly emphasize atmospheric traces of lithium and other individual elements as signifiers of the types of rock found on Earth. The studies don’t adequately account for different minerals made up of various ratios of those elements, and the possibility of rock types not found on Earth that contain those minerals.

Based on the texts, how would Putirka and Xu (Text 2) most likely characterize the conclusion presented in Text 1?

- A. As unexpected, because it was widely believed at the time that white dwarf exoplanets lack continental crusts
- B. As premature, because researchers have only just begun trying to determine what kinds of crusts white dwarf exoplanets had
- C. As questionable, because it rests on an incomplete consideration of potential sources of the elements detected in white dwarf atmospheres

- D. As puzzling, because it's unusual to successfully detect lithium and sodium when analyzing wavelengths of light in white dwarf atmospheres

Asteroid 6478 Gault has experienced intermittent mass loss since at least 2013, but in contrast to some other asteroids with repeated mass-loss episodes, 6478 Gault has not lost mass at its perihelion (the closest point of its orbit to the Sun), and thus the loss is not attributable to solar energy–driven ice vaporization. And as Jane X. Luu et al. point out, the singular nature of impact ejection makes it untenable as an account of multiple loss episodes of similar duration over several years. Instead, Luu et al. are likely correct that 6478 Gault is shedding mass due to rotational instability.

Which choice best describes the overall structure of the text?

- A. It presents a scientific observation, describes a contrast between that observation and other observations, and then explains why those other observations should not be considered credible.
- B. It describes an astronomical finding, discusses competing theories about that finding that the author regards as flawed, and then describes new evidence that supports an alternative theory.
- C. It introduces a natural phenomenon, refutes two potential explanations for that phenomenon, and then presents a third explanation for that phenomenon that the author regards as plausible.
- D. It discusses a physical process, evaluates possible causes of that process, and then states that a persuasive account of the process has yet to be put forward.

The province of Xoconochco was situated on the Pacific coast, hundreds of kilometers southeast of Tenochtitlan, the capital of the Aztec Empire. Because Xoconochco's location within the empire was so _____ blank, cacao and other trade goods produced there could reach the capital only after a long overland journey.

Which choice completes the text with the most logical and precise word or phrase?

- A. unobtrusive
- B. concealed
- C. approximate
- D. peripheral

Text 1

Growth in the use of novel nanohybrids—materials created from the conjugation of multiple distinct nanomaterials, such as iron oxide and gold nanomaterials conjugated for use in magnetic imaging—has outpaced studies of nanohybrids’ environmental risks. Unfortunately, risk evaluations based on nanohybrids’ constituents are not reliable: conjugation may alter constituents’ physiochemical properties such that innocuous nanomaterials form a nanohybrid that is anything but.

Text 2

The potential for enhanced toxicity of nanohybrids relative to the toxicity of constituent nanomaterials has drawn deserved attention, but the effects of nanomaterial conjugation vary by case. For instance, it was recently shown that a nanohybrid of silicon dioxide and zinc oxide preserved the desired optical transparency of zinc oxide nanoparticles while mitigating the nanoparticles’ potential to damage DNA.

Based on the texts, how would the author of Text 2 most likely respond to the assertion in the underlined portion of Text 1?

- A. By concurring that the risk described in Text 1 should be evaluated but emphasizing that the risk is more than offset by the potential benefits of nanomaterial conjugation
- B. By arguing that the situation described in Text 1 may not be representative but conceding that the effects of nanomaterial conjugation are harder to predict than researchers had expected
- C. By denying that the circumstance described in Text 1 is likely to occur but acknowledging that many aspects of nanomaterial conjugation are still poorly understood

- D. By agreeing that the possibility described in Text 1 is a cause for concern but pointing out that nanomaterial conjugation does not inevitably produce that result

Scholarly accounts of the Chicano movement—a movement that advocated for the social, political, and cultural empowerment of Mexican Americans and reached its zenith in the 1960s and 1970s—tend to focus on the most militant, outspoken figures in the movement, making it seem uniformly radical. Geographer Juan Herrera has shown, however, that if we shift our focus toward the way the movement manifested in comparatively low-profile neighborhood institutions and projects, we see participants espousing an array of political orientations and approaches to community activism.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It presents a trend in scholarship on the Chicano movement that the text claims has been reevaluated by researchers in light of Herrera’s work on the movement’s participants.
- B. It identifies an aspect of the Chicano movement that the text implies was overemphasized by scholars due to their own political orientations.
- C. It describes a common approach to studying the Chicano movement that, according to the text, obscures the ideological diversity of the movement’s participants.
- D. It summarizes the conventional method for analyzing the Chicano movement, which the text suggests creates a misleading impression of the effectiveness of neighborhood institutions and projects.